SOP - Troubleshooting QoS Congestion Event on Ericsson PE (ELEC-PE-06 at Electronic City Hub)

Node: ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_5279

Description

This Standard Operating Procedure (SOP) outlines the step-by-step diagnostic and resolution procedures for addressing QoS congestion events on the Ericsson PE (Packet Edge) device, specifically on Node ID ELEC-PE-06 located at the Electronic City Hub. The failure domain is characterized by microbursts, interface tail drops, and QoS queue starvation, as indicated by the ALARM_5279 alarm code of critical severity. The underlying cause of this failure domain is primarily attributed to the interaction between the packet forwarding engine, the queueing scheduler, and the congestion avoidance mechanisms implemented on the PE device. When the PE device experiences a surge in packet arrival rates or high-traffic loads, it may lead to queueing congestion, resulting in packet drops and subsequent alarms. In order to mitigate this issue, it is essential to ensure proper configuration of QoS policies, optimal queue sizing, and adequate buffer allocation. Additionally, real-time monitoring of traffic patterns and queue utilization is crucial for proactive identification and resolution of potential congestion hotspots.

Pre-requisite Checks

- Verify that the Electronic City Hub is accessible and remotely manageable via the Ericsson PE device's CLI.

- Confirm that all redundant links and backup power supplies are operational and not contributing to the congestion event.

- Ensure that the PE device's hardware and software configurations are up-to-date, with the latest firmware and software patches applied.

Potential Root Causes

Cause	Probability
Inadequate QoS policy configuration	High
Incorrect queue sizing or buffer allocation	Medium
Congestion avoidance mechanisms misconfigured or disabled	Low

Diagnostic Commands & Expected Outputs

```
Command: show clock, Expected Output: System clock is synchronized with the NTP server (e.g., 2023-12-31 23:59:59 UTC), Description: Verifies the system clock is accurately set, ensuring timestamping of all log events and performance metrics.
```

```
Command: show qos policy, Expected Output: QoS policy configuration (e.g., policing, queuing, and shaping) is correctly applied to all interfaces (e.g., interface Ethernet 0/0/0), Description: Examines the QoS policy configuration to identify any potential misconfigurations or inconsistencies.

Command: show interface statistics, Expected Output: Interface statistics show no signs of congestion (e.g., no drops, no overflows), Description: Provides a snapshot of real-time interface statistics, enabling the identification of potential congestion hotspots.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policy configuration, Command: configure terminal policy-map class bandwidth , Description: Re-examines the QoS policy configuration and adjusts it as necessary to ensure proper policing, queuing, and shaping of traffic.

Step 2: Step: Reconfigure queue sizing and buffer allocation, Command: configure terminal interface Ethernet 0/0/0 queue-size , Description: Reconfigures the queue size and buffer allocation on the affected interface to accommodate increased traffic loads.

Step 3: Step: Re-enable congestion avoidance mechanisms, Command: configure terminal congestion-avoidance , Description: Re-enables congestion avoidance mechanisms, such as random early detection (RED), to prevent queue buildup and subsequent packet drops.

Post-Check Verification

```
Command: show qos policy, Expected Output: QoS policy configuration is correctly applied to all interfaces (e.g., interface Ethernet 0/0/0), Description: Verifies that the QoS policy configuration has been correctly adjusted and applied to all interfaces.

Command: show interface statistics, Expected Output: Interface statistics show no signs of congestion (e.g., no drops, no overflows), Description: Verifies that the interface statistics are within normal ranges, indicating the successful resolution of the congestion event.
```

Roll-back Procedures

- Step: Restore previous QoS policy configuration, Command: configure terminal rollback , Description: Restores the previous QoS policy configuration in case the adjustment made during resolution steps failed to resolve the issue.

- Step: Restore previous queue sizing and buffer allocation, Command: configure terminal interface Ethernet 0/0/0 rollback queue-size, Description: Restores the previous queue size and buffer allocation configuration in case the adjustment made during resolution steps failed to resolve the issue.

- Step: Disable congestion avoidance mechanisms, Command: configure terminal configuration-default , Description: Disables congestion avoidance mechanisms, such as random early detection (RED), in case the re-enablement made during resolution steps failed to resolve the issue.